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Escalation of mechanical circulatory support in patients with STEMI-related cardiogenic shock: a *post hoc* analysis of the DanGer Shock trial

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Aims

In STEMI-related cardiogenic shock, left ventricular unloading with a micro-axial flow pump (mAFP) improved survival in the Danish German (DanGer) Shock trial, though some patients in both groups required escalation to higher volume temporary mechanical circulatory support systems (tMCS). This study aims to investigate tMCS escalation strategies and outcome in non-comatose patients with STEMI-related cardiogenic shock.

Methods and results

In this *post hoc* analysis of the DanGer Shock trial, patients in both study groups were stratified as escalated vs. not escalated to treatments other than a percutaneously implanted mAFP (Impella CP®). Logistic regression models were fitted to investigate the differential association of baseline markers with the likelihood of an escalation in patients randomized to mAFP vs. standard of care, adjusted for age and country. The 180-day all-cause mortality risk for the four groups was evaluated using the Kaplan–Meier method. Of the 355 patients included in the DanGer Shock trial, 60 underwent tMCS escalation, 25/179 (14%) allocated to the mAFP group, and 35/176 (20%) allocated to the control group (adjusted odds ratio 0.52, 95% confidence interval 0.30–0.91). Median time from randomization to escalation was 12 h (2.7–66.5) in the mAFP arm vs. 0.7 h (IQR: 0.4–2.7) in the standard of care arm ($P < 0.001$). A higher baseline lactate predicted an escalation in the standard of care arm (odds ratio 1.80, 95% confidence interval 1.14–3.13), but not in the mAFP arm (odds ratio 0.57, 95% confidence

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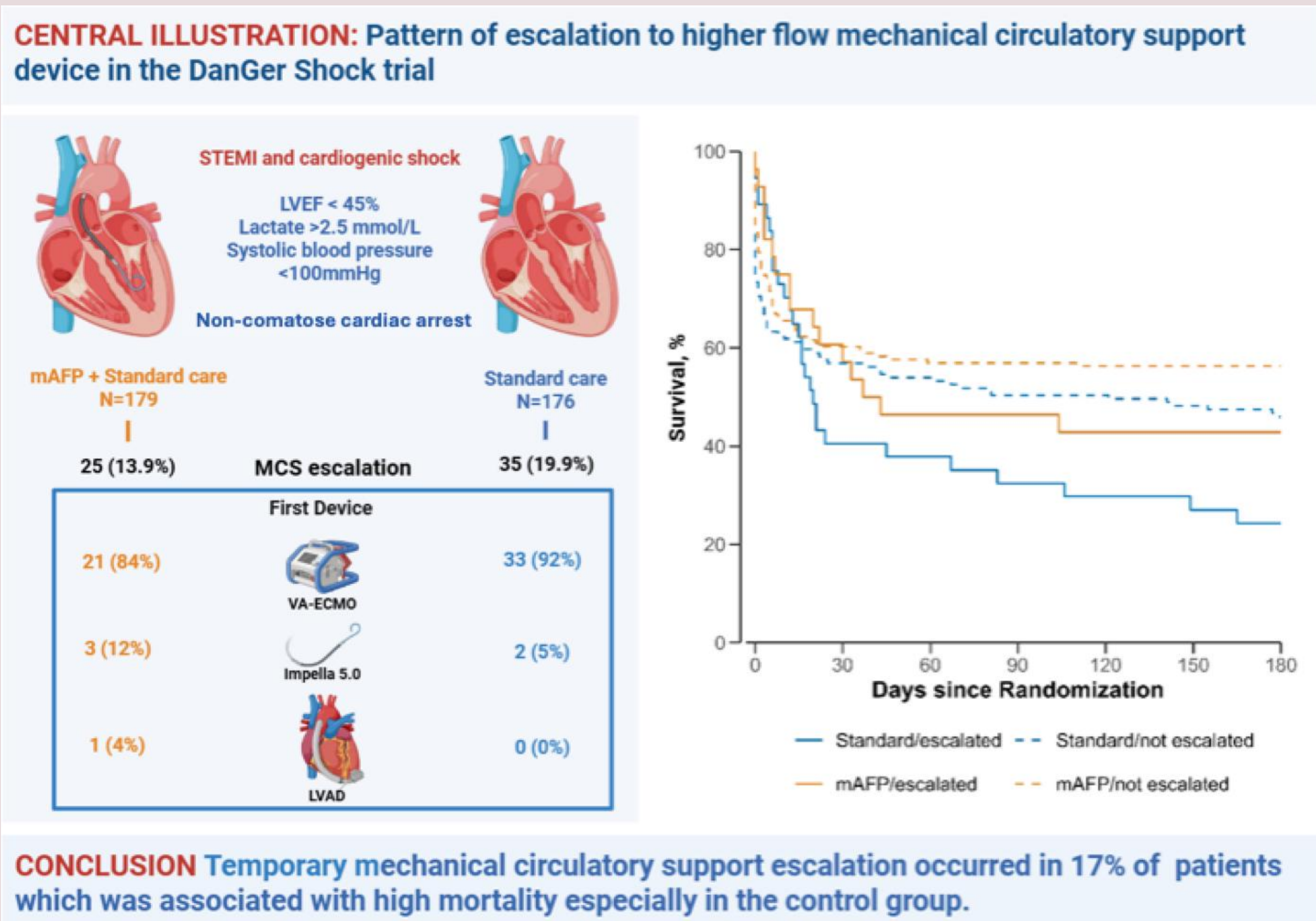
interval 0.29–1.15, interaction-$P < 0.01$). Mortality was significantly higher in patients who underwent tMCS escalation (71.7% vs. 48.1%, $P < 0.01$) and highest in escalated patients from the standard of care arm (80.0%). Interestingly, those escalated in the mAFP arm had a comparable mortality risk to the non-escalated patients in the standard of care group (60.0% vs. 53.2%).

Conclusion In the DanGer Shock trial of STEMI-related cardiogenic shock, escalation to tMCS occurred in almost 1/5 patients, was more likely in the control group, and was associated with a higher mortality risk. Even with early escalation (<1 h), patients escalated from the control group had the highest mortality, underscoring the need for future trials to define optimal escalation strategies.

Graphical Abstract



mAFP, micro-axial flow pump; ECMO, veno-arterial extracorporeal membrane oxygenation therapy; LVAD, emergency implantation of a left ventricular assist device; LVEF, left ventricular ejection fraction.

Keywords Cardiogenic shock • Escalation • Mechanical circulatory support • mAFP

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Introduction

The DanGer Shock trial established evidence for applying micro-axial flow pumps (mAFP) in selected non-comatose patients with STEMI-related cardiogenic shock.[1] Subsequently, clinical guidelines in the USA have implemented a 2A recommendation for mAFP for this indication.[2]

In some patients with severe STEMI-related cardiogenic shock, treatment with a percutaneously implantable mAFP alone may not suffice: either because of severe cardiogenic shock with the need for cardiac output support exceeding the capabilities of the percutaneously implantable mAFP, or because of concomitant right ventricular failure or pulmonary failure with inadequate mAFP performance.[3] Veno-arterial extracorporeal membrane oxygenation

(VA-ECMO) as well as surgically implantable mAFP (e.g. Impella 5.0/5.5®) can provide higher systemic flow than the percutaneously implantable mAFP, and the former also allows for right ventricular support as well as gas exchange, making it the treatment of choice in these scenarios. However, these escalation options all have drawbacks: VA-ECMO carries a higher risk profile than the mAFP, comes with certain haemodynamic limitations, and is overall a more complex and resource demanding treatment.[4,5] Surgically implantable mAFP require specialized personnel and operating resources, which limits its applicability in the emergency setting. Ultimately, optimal treatment strategies for these patients, specifically when and how to escalate to more potent temporary mechanical circulatory support devices (tMCS), remain unclear.

The aim of this study was to investigate escalation pathways in the prospective, randomized DanGer Shock trial.

Methods

Study design

This was an exploratory *post hoc* analysis of the DanGer Shock Trial, which was an international, multicentre, open-label, randomized clinical trial conducted in Denmark, Germany, and the UK, investigating the efficacy of a mAFP (Impella CP®, Abiomed) in addition to standard of care on 180-day mortality in patients with STEMI-related cardiogenic shock. The trial protocol and primary results have been published previously, and the trial was registered at www.clinicaltrials.gov (NCT01633502) and approved by the ethics committee at each participating site.[1,6] The data are available upon reasonable request to the authors.

In brief, from 2013 until 2023, non-comatose patients with STEMI-related cardiogenic shock were randomized 1:1 to either treatment with a mAFP on top of standard of care or standard of care alone. Randomization was possible until 12 h after the index percutaneous coronary intervention, with immediate implantation of the mAFP in case of randomization to the mAFP group. Within this setting, escalation to more potent devices was possible whenever needed, and the protocol specifically recommended tMCS escalation to be considered in case of

(1) *Critical cardiogenic shock* defined as cardiac index < 2 L/min/m^2, mean arterial blood pressure < 60 mmHg, and sustained/worsening acidosis and lactate levels despite optimized inotropic support (adrenaline, milrinone, norepinephrine, dobutamine), optimized volume status (CVP 10–15 mmHg), and continuous renal replacement therapy if indicated and Impella device if allocated.
(2) *Progressive decline* despite circulatory support defined as continuous need for inotropic support/Impella support if allocated more than 48 h after randomization (inability to wean from circulatory support) and signs of progressive organ hypoperfusion with increasing lactate that is not caused by severe sepsis (see protocol for definition).
(3) *Stable but dependent* on circulatory support/inotropes defined as patients that 5 days after initiation of circulatory support are stable but unable to be weaned from circulatory support and/or managed with ≥1 inotropic agent, not due to severe sepsis.

The final decision of escalation was at the discretion of the local shock team in both study arms.

For this *post hoc* analysis, patients were stratified as escalated vs. not escalated based on any use of a VA-ECMO, a surgically implanted mAFP or emergency surgical implantation of a left ventricular assist device (LVAD). Patients receiving an elective LVAD implantation ($n = 14$) were not considered as tMCS escalation. Cross-over from the control arm to implantation of mAFP (Impella CP) was not considered as an escalation in accordance with the main trial.

Consistent with the main publication, 180-day all-cause mortality was the primary endpoint for this analysis.

Statistical analysis

Categorical variables were presented as numbers and percentages and analysed using the χ^2 or Fisher's exact test, whereas continuous variables were presented as median [interquartile range (IQR)] and using the Wilcoxon rank-sum test.

To investigate predictors of tMCS escalation in the presence of competing risks, we used a semiparametric proportional odds model for the cumulative incidence function. Escalation was treated as the event of interest, with death and hospital discharge without escalation treated as competing events. Time was defined as time to the first escalation, death, or discharge alive from the hospital. The model included several clinically selected predictors as independent variables, adjusted for age and country of the treating hospital. Additionally, an interaction term for randomization to the mAFP arm vs. the control arm was assessed to evaluate whether the associations between the clinical predictors and escalation differed between study arms.

The Kaplan–Meier method was used to assess 180-day all-cause mortality in patients escalated vs. not escalated in both study arms. A landmark analysis, excluding patients who died on the first admission day and assessing escalation only in patients who were escalated within 24 h, was conducted to reduce the influence of immortal time bias, i.e. patients who died before being able to be escalated.

Additionally, non-escalated patients who died within 72 h after admission to the intensive care unit and those who were discharged from the intensive care units within 72 h after admission were descriptively analysed as a sensitivity analysis to get a sense of characteristics of patients deemed too sick or ineligible for escalation or too well to be escalated.

The *P*-values were two-sided and considered statistically significant if <0.05. All statistical analyses were performed in R Studio, version 4.5.0 [RStudio Team (2020). RStudio: Integrated Development for R. RStudio, PBC, Boston, MA; URL: http://www.rstudio.com/] using extension packages *survival and timereg*.

Results

Study cohort

All 355 patients from the intention-to-treat population of the main analysis were considered for this *post hoc* analysis. Of these, 60 patients (17%) were escalated to a tMCS device, 25 of 179 patients (14%) in the mAFP arm, and 35 of 176 patients (20%) in the control group. First device was a VA-ECMO in 54 (90.0%), Impella 5.0 in 5 (8.3%), and an emergency LVAD implantation in one patient (1.7%) (*Graphical Abstract*).

Patients escalated were younger, had a lower left ventricular ejection fraction at baseline, more commonly presented with a culprit lesion in the left main artery, and were less frequently resuscitated or intubated before randomization. Similar trends were seen when patients escalated vs. not escalated were further stratified based on their allocated study arm (*Table 1*; Supplementary material online, *Tables S1* and *S2*). Escalation was also numerically more frequent among male patients, although this did not reach statistical significance.

Median time from randomization to tMCS escalation was overall 1.9 h (IQR: 0.65–22.2) [mAFP group: 12 h (2.7–66.5) and 0.7 h (IQR: 0.4–2.7) in the control group, $P < 0.001$]. In the standard of care group, device escalation to one device was done in 28 patients and escalation to two temporary MCS systems in 6 patients. In the mAFP group, one additional temporary MCS device was placed in 21 patients and two devices in 4 patients. Durable LVAD was implanted in three patients escalated to tMCS (8.6%) in standard of care and 6 (24.0%) in the mAFP

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Table 1 Baseline characteristics of the overall study cohort

Characteristic	Overall n = 355	Escalated n = 60	Not escalated n = 295	*P*-value
Allocated to mAFP, no. (%)	179 (50%)	25 (42%)	154 (52%)	0.14
Demographics				
Age, median (IQR)	69 (59, 76)	64 (56, 70)	70 (60, 77)	<0.001
Sex, *n* (%)				0.061
Female	74 (21%)	8 (12%)	67 (23%)	
Male	281 (79%)	53 (88%)	228 (77%)	
Medical history				
Hypertension, no. (%)	183 (52%)	31 (52%)	152 (52%)	0.98
Diabetes, no. (%)	80 (23%)	12 (20%)	68 (23%)	0.61
Myocardial infarction, no. (%)	57 (16%)	5 (8.3%)	52 (18%)	0.074
Heart failure, no. (%)	28 (7.9%)	7 (12%)	21 (7.1%)	0.43
Chronic kidney disease, no. (%)	35 (9.9%)	3 (5.0%)	32 (11%)	0.17
Clinical presentation				
Systolic blood pressure at randomization, median (IQR)	82 (72, 91)	80 (70, 91)	83 (73, 91)	0.45
Mean arterial blood pressure at randomization, median (IQR)	63 (55, 72)	64 (55, 73)	63 (55, 71)	0.75
Heart rate at randomization, median (IQR)	95 (77, 110)	96 (77, 112)	95 (77, 110)	0.66
Lactate concentration at randomization, median (IQR)	4.5 (3.3, 7.0)	4.4 (3.3, 8.1)	4.5 (3.3, 7.0)	0.96
pH at randomization, median (IQR)	7.29 (7.17, 7.37)	7.28 (7.18, 7.35)	7.29 (7.17, 7.37)	0.86
LVEF at randomization (%), median (IQR)	25 (15, 30)	20 (10, 29)	25 (15, 30)	0.003
Resuscitation before randomization, no. (%)	72 (20%)	6 (10%)	66 (22%)	0.030
Endotracheal intubation before arrival at cath lab, no. (%)	63 (18%)	6 (10%)	57 (19%)	0.085
Anterior myocardial infarction, no. (%)	255 (72%)	49 (82%)	206 (70%)	0.063
SCAI stage at admission, *n* (%)				0.78
C	197 (55%)	31 (52%)	166 (56%)	
D	101 (28%)	18 (30%)	83 (28%)	
E	57 (16%)	11 (18%)	46 (16%)	
Procedural characteristics				
PCI of culprit, no. (%)	343 (97%)	59 (98%)	284 (96%)	0.70
Multivessel disease on CAG, no. (%)	256 (72%)	49 (82%)	207 (70%)	0.070
Time from symptom onset to balloon inflation, minutes, median (IQR)	242 (138, 644)	239 (142, 490)	244 (137, 697)	0.73
Culprit lesion, *n* (%)				0.004
LAD	196 (56%)	29 (48%)	167 (57%)	
LCX	40 (11%)	5 (8.3%)	35 (12%)	
RCA	55 (16%)	5 (8.3%)	50 (17%)	
LM	59 (17%)	20 (33%)	39 (13%)	
Graft	3 (0.8%)	1 (1.7%)	2 (0.7%)	

mAFP, micro-axial flow pump; IQR, interquartile range; LVEF, left ventricular ejection fraction; PCI, percutaneous coronary intervention; LAD, left anterior descending artery; LCX, left circumflex artery; RCA, right coronary artery; LM, left main artery.

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group while on tMCS. Other variables depicting in-hospital course are shown in *Table 2*.

Among the non-escalated patients, n = 83 patients deceased within 72 h after admission to the intensive care unit whereas n = 75 patients were discharged within 72 h after admission. In comparison, non-escalated patients with an early discharge were significantly younger and had a higher blood pressure and lower heart rate, a lower lactate, and a higher ejection fraction (see Supplementary material online, *Table S3*).

Predictors of escalation

In the overall study group, higher age, higher left ventricular ejection fraction at baseline, resuscitation before arrival to the catheter room, and randomization to the mAFP group were significantly associated with a lower likelihood of being escalated, whereas a culprit lesion in the left main coronary artery was significantly associated with a higher likelihood of being escalated, accounting for competing risk of death and discharge and adjustment for age and country of the treating hospital (*Figure 1*).

Table 2 In-hospital course of the overall study cohort

Characteristic	Overall *n* = 355	Escalated *n* = 60	Not escalated *n* = 295	*P*-value
Duration of Impella support (hours), median (IQR)	61 (31, 95)	122 (45, 309)	58 (30, 82)	0.001
Mechanical haemolysis, no. (%)	23 (12%)	3 (8.8%)	20 (12%)	0.77
Device malfunction, no. (%)	3 (1.5%)	2 (5.4%)	1 (0.6%)	0.090
Mechanical ventilation, no. (%)	250 (70%)	57 (95%)	193 (65%)	<0.001
Duration of mechanical ventilation (days), median (IQR)	4 (1, 10)	10 (6, 21)	3 (1, 8)	<0.001
Any vasopressor, no. (%)	305 (86%)	59 (98%)	246 (83%)	0.002
Norepinephrine infusion during ICU treatment, no. (%)	298 (84%)	58 (97%)	240 (81%)	0.003
Dopamine infusion during ICU treatment, no. (%)	92 (26%)	14 (23%)	78 (26%)	0.62
Epinephrine infusion during ICU treatment, no. (%)	133 (37%)	36 (60%)	97 (33%)	<0.001
Any inotrope, no. (%)	233 (66%)	55 (92%)	178 (60%)	<0.001
Dobutamine infusion during ICU treatment, no. (%)	121 (34%)	30 (50%)	91 (31%)	0.004
Milrinone infusion during ICU treatment, no. (%)	121 (34%)	30 (50%)	91 (31%)	0.004
Levosimendan infusion during ICU treatment, no. (%)	79 (22%)	24 (40%)	55 (19%)	<0.001
Additional PCI during hospitalization, no. (%)	17 (4.8%)	3 (5.0%)	14 (4.7%)	>0.99
Additional CABG during hospitalization, no. (%)	3 (0.8%)	1 (1.7%)	2 (0.7%)	0.43
Median duration of ICU admission, days, median (IQR)	4 (1, 12)	13 (6, 27)	3 (0, 9)	<0.001
Still in ICU at Day 30, no. (%)	34 (9.6%)	14 (23%)	20 (6.8%)	<0.001
Median duration of hospitalization, days, median (IQR)	9 (2, 21)	20 (7, 36)	7 (2, 20)	<0.001
Still in hospital at Day 30, no. (%)	65 (18%)	20 (33%)	45 (15%)	<0.001

IQR, interquartile range; ICU, intensive care unit; PCI, percutaneous coronary intervention; CABG, coronary artery bypass grafting.

Most of those associations were persistent when the analysis was stratified by treatment allocation. However, there was a differential association for lactate at baseline, as a higher lactate was significantly associated with a higher likelihood of being escalated in the control group (odds ratio 1.89, 95% confidence interval 1.14–3.13), but not in the mAFP group (odds ratio 0.57, 95% confidence interval 0.29–1.16, interaction-$P < 0.01$). Also, there was a trend towards a differential association for age and CS severity, indicating a lower likelihood for escalation in older patients allocated to the standard of care group, but not to the mAFP arm (interaction-*P* 0.07) and a higher likelihood for escalation in patients with SCAI class D/E in the standard of care group, but not in the mAFP group (interaction-*P* 0.09).

Outcome

Overall, 180-day all-cause mortality in the study cohort was 52.1%. Escalated patients had a significantly higher mortality rate than non-escalated patients (71.7% vs. 48.1%, $P < 0.01$). When considering randomized study group, mortality was highest in escalated patients from the control arm (28 of 35 patients, 80.0%), whereas mortality in escalated patients in the mAFP group and non-escalated patients in the control group was 15 of 25, 60.0% and 75 of 141, 53.2%, respectively. The lowest mortality was found in non-escalated patients from the mAFP group (67 of 154, 43.5%).

The corresponding Kaplan–Meier curves indicated that the increase of mortality risk seen in the trial from Day 30 until Day 180 was especially present in escalated patients from both study groups (*Figure 2*). Supplementary material online, *Table S4* shows causes of death, and Supplementary material online, *Table S5* adverse events across the four groups.

Discussion

In this *post hoc* analysis of the DanGer Shock trial, 17% of all enrolled patients were escalated to a higher flow tMCS device. Several variables related to cardiogenic shock or STEMI severity as well as allocation to mAFP were predictive of treatment escalation in the overall study cohort. However, only baseline lactate showed a differential association, where higher lactate values were only predictive for escalation in the control group, but not in the mAFP group. Importantly, despite a median time to escalation of <1 h, mortality risk was highest in escalated patients from the control group.

The DanGer Shock trial was the first positive trial of an immediate tMCS strategy in the setting of cardiogenic shock, confirming the hypothesis that targeted support for the left ventricle in carefully selected, non-comatose patients with STEMI-related cardiogenic shock reduces mortality at 180 days.[1] However, cardiogenic shock is a heterogeneous and sometimes quickly deteriorating disease, so that even in these carefully selected patients sole support with a mAFP may not always suffice.[7] Reasons for this range from overt left ventricular failure requiring higher flow than the percutaneously implanted mAFP can provide, malrotation and poor device function, to concomitant respiratory failure or biventricular failure, which cannot be directly addressed by the percutaneously implanted mAFP. As a more widespread use of the percutaneously implanted mAFP may follow the results of the DanGer Shock trial and the positive guideline recommendation, data on the optimal treatment strategies for patients with severe cardiogenic shock who deteriorate despite mAFP treatment is utterly needed.

In the light of this, the lingering questions is whether the DanGer Shock strategy, i.e. upfront use of a percutaneously

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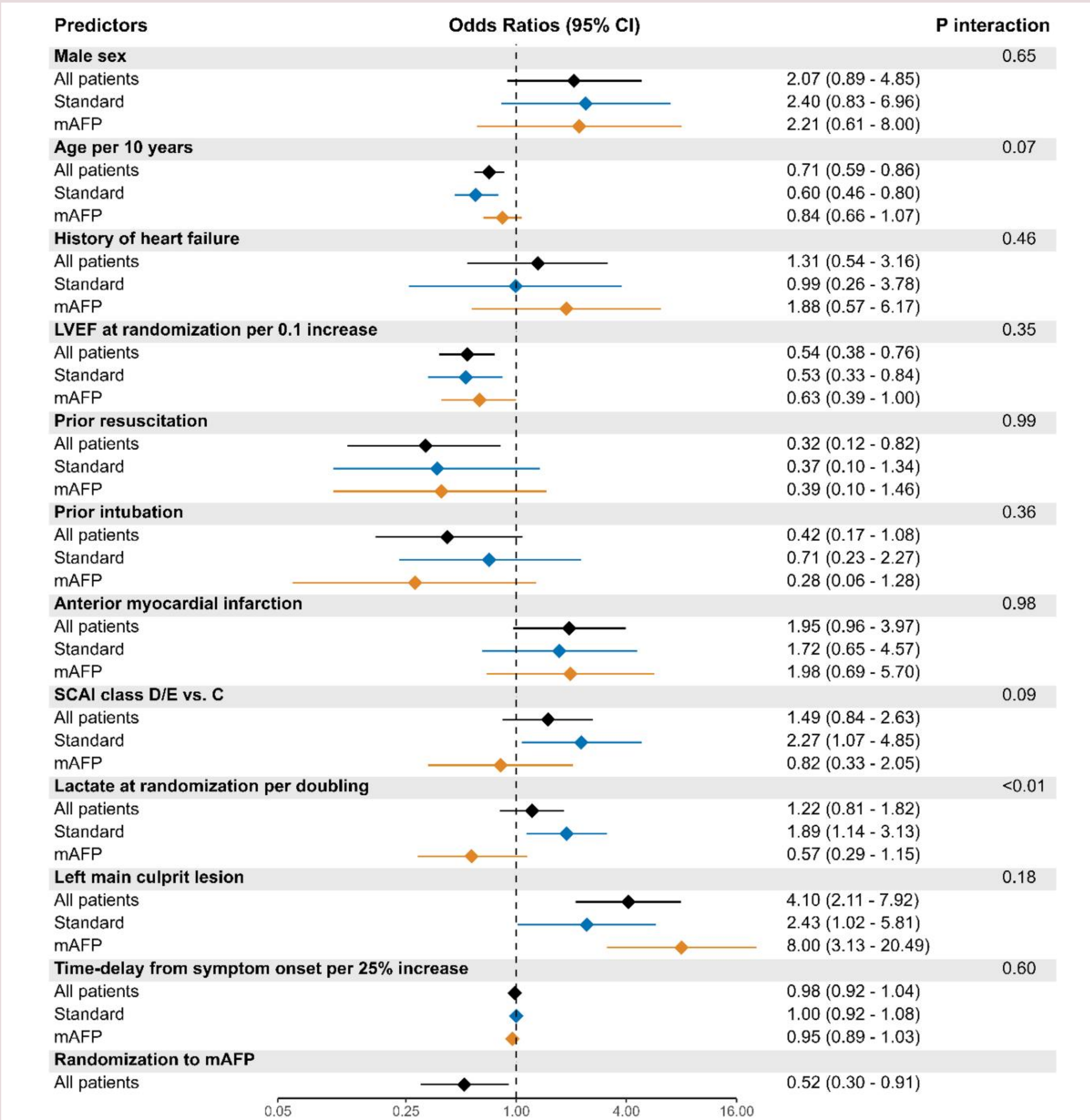

Figure 1 Predictors of escalation. Association between selected clinical predictors and escalation therapy, shown for all patients and stratified by treatment arm (standard or micro-axial flow pump). Estimates come from competing risk models that account for death or discharge without escalation. All models adjust for age and country. The *P* interaction indicates whether the association with escalation differed significantly between the two treatment arms. mAFP, micro-axial flow pump; LVEF, left-ventricular ejection fraction; OR, odds ratio; CI, confidence interval.

implanted mAFP, also translates to patients at risk of or with deteriorating cardiogenic shock, or if these patients should directly be implanted with a more potent tMCS device. Prior registry studies with uncontrollable bias have supported the latter, showing an association between earlier use of VA-ECMO and lower mortality risk as compared to a delayed application[8,9]; and similarly, a lower mortality risk has been reported if the more potent, surgically implanted mAFP is used as the primary treatment strategy.[10] The pathophysiological rational for this is appealing, an early intervention to support the failing heart and to stop the vicious cycle of cardiogenic shock, thereby preventing further end-organ damage and ultimately preventing death.[11] However, there are also studies that contradict this hypothesis: although small, the prospective randomized ECMO-CS

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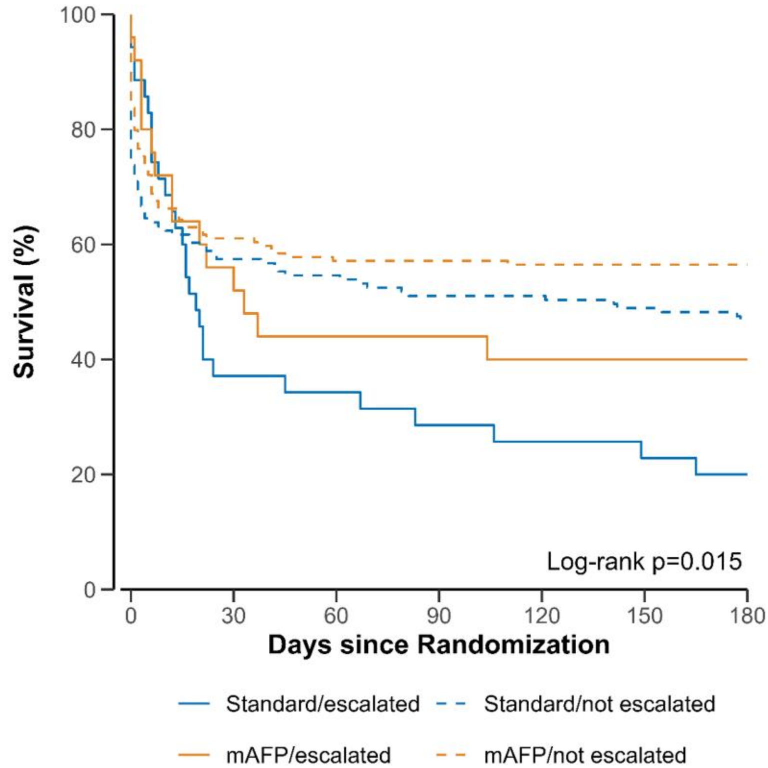

No. at Risk							
Standard/escalated	35	13	12	10	9	8	7
mAFP/escalated	25	14	11	11	10	10	10
Standard/not escalated	141	81	77	72	72	69	65
mAFP/not escalated	154	94	88	88	87	87	87

Figure 2 Kaplan–Meier curves for the primary endpoint of 180-day all-cause mortality. Solid lines represent patients who required escalation; dashed lines represent those who did not escalate. Median time from randomization to escalation was 1.9 h (interquartile range: 0.65–22.2) [micro-axial flow pump: 12 h (2.7–66.5) and 0.7 h (interquartile range: 0.4–2.7)]; thus, most patients were escalated at Day 0. Because patients had to survive long enough to receive escalation, comparisons between escalated and non-escalated strata may be influenced by immortal-time bias; survival curves should therefore be interpreted descriptively. mAFP, micro-axial flow pump.

study has shown no survival benefit with an early vs. a delayed use of VA-ECMO in patients with deteriorating cardiogenic shock.[12] Additionally, a large multicentre registry has similarly shown that a delayed use of VA-ECMO is associated with similar outcomes than an early use of VA-ECMO.[13] The idea behind this concept is that a delayed use provides an even better selection for these highly invasive devices, as patients who are about to stabilize with the established treatments are spared from the potential risk of complications.[3]

In this context, the findings of this *post hoc* analysis of the DanGer Shock trial may support a mAFP first approach: although roughly one in five patients was in need of an escalation treatment, mortality of these patients in the mAFP group (e.g. primarily randomized to and treated with a percutaneously implanted mAFP, then escalated to a more potent device) was lower as compared to escalated patients in the control arm and even comparable to the non-escalated patients in the control arm. This was observed despite a median time to escalation of <1 h in the control group, indicating that a similar benefit may not be achieved with early VA-ECMO escalation in selected patients with SCAI stage D or E shock. Hence, a mAFP first strategy seems to preserve the benefit of an early left ventricular unloading even in deteriorating patients. This is in line with retrospective studies which have supported the upfront use of left ventricular unloading in patients with severe cardiogenic shock in need of VA-ECMO support.[14,15] Importantly, upfront escalation always needs to be considered in the context of the given case based on shock team consensus. In this study, a relevant proportion of patients in the non-escalated group deceased within 72 h of admission to the intensive care unit, indicating that the local shock team either saw a contraindication to escalation or determined it to be futile. Future randomized studies such as the ongoing UNLOAD ECMO trial (NCT05577195) are needed to confirm this approach and provide evidence also for deteriorating patients with severe CS.

Importantly, although this study is based on one of the largest randomized trials on cardiogenic shock, with complete data capture regarding escalation treatment and complete follow-up of enrolled patients, its results may need to be interpreted with caution. The primary limitation is certainly the retrospective

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post hoc nature of the analysis, so that unknown or unmeasured confounding cannot be ruled out. Decision on escalation was protocolized but left to the discretion of the shock team but did also reflect the local practice which may differ in other geographic regions and also may have changed over the 10-year period of the trial. Also, the reasons for escalation were not captured in the database and can therefore not be analysed in the context of this study. Differences in timing of escalation (early vs. late) likely reflect different reasons for escalation in the two groups, but this was not considered in the analyses to avoid an immortal time bias. Although the overall median time to escalation was <2 h, this limitation needs to be recognized for when interpreting the results. Analysing sex-specific reasons to abstain from escalation (such as anticipated bleeding with small femoral vessel diameter or anticipated mAFP malpositioning with small left ventricular cavities) would be of particular interest, as we observed a trend towards a higher frequency of escalation among male patients. The short median time to escalation in the control group is also likely to indicate severe shock that did not immediately respond to therapy, rather than deterioration of established therapy, which needs to be considered when interpreting the results. DanGer Shock enrolled patients over a decade during which the use of tMCS evolved. The concept of combining VA-ECMO and mAFP did not exist and the use of VA-ECMO as rescue management of AMI-CS and refractory cardiac arrest was not generally implemented. Also, while the data are extremely valid for enrolled patients, generalizability to other patients and especially centres not participating in the trial might be limited. Upcoming randomized trials enrolling patients with severe cardiogenic shock treated with left ventricular unload devices on top of VA-ECMO, such as UNLOAD ECMO (NCT: NCT05577195), ANCHOR (NCT: NCT04184635), or REMAP ECMO (NCT: NCT06522594), and upcoming trials testing a direct intervention with a more potent mAFP, will provide further and potentially definitive insights into this important topic.

Conclusions

In the DanGer Shock trial of STEMI-related cardiogenic shock, escalation to tMCS occurred in almost 1/5 patients, was more likely in the control group, and was associated with a higher mortality risk. Even with early escalation (<1 h), patients escalated from the control group had the highest mortality, underscoring the need for future trials to define optimal escalation strategies.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

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